Singha Junchan

Cloud-Native Security · Authorization and Access Control · Verifiable Audit · AI-Agent Security

singhaju.github.io | github.com/singhaju | linkedin.com/in/singha-junchan | Pathum Thani, Thailand

Research Interests

Security of cloud-native systems: **zero-trust authorization and access control** for Kubernetes and multi-cloud systems, policy compilation, **verifiable audit** with zero-knowledge proofs and Merkle commitments, **post-quantum authorization**, and **security for AI agents**. My working idea: *a security check should be cheap enough to run on every request and provable afterwards.* I build working prototypes on real Kubernetes and AWS and measure them against published baselines.

Education

B.Eng. in Computer Engineering Aug 2023 – Expected May 2027

Sirindhorn International Institute of Technology (SIIT), Thammasat University *Pathum Thani, Thailand*

- **GPA 3.67 / 4.00**, first-class honours expected; **minor in Cloud Computing & Cybersecurity**.
- **Full scholarship**: 4-Year Continuing Academic Outstanding Student Program.
- Research supervisor: **Prof. Somchart Fugkeaw**, School of ICT, SIIT.

Publications

2 accepted (one Best Paper Award) · **3 submitted** · **4 in preparation**. First author of all except U-HAP (second author).

Accepted

U-HAP: Unified Authorization over Heterogeneous Policies with Efficient Multi-Resource Verification in Kubernetes **Best Paper Award** · IEEE JCSSE 2026

K. Jairak, **S. Junchan**, P. Pruksorranan, S. Fugkeaw

23rd International Joint Conference on Computer Science and Software Engineering, Bangkok, June 2026; I presented the paper. ACL, RBAC, ABAC and deny policies are compiled into indexes for a Kubernetes webhook authorizer: policy updates **1.6–2.7× faster than Open Policy Agent** with **100/100 identical decisions**, and up to 16.9× faster ABAC evaluation at 320 rules. Pages 79–84, DOI 10.1109/JCSSE68839.2026.11597088.

ZKAudit: Scalable Zero-Knowledge Auditor Authentication and Fast Integrity Verification for Cloud Logs **Accepted** · KST 2027

S. Junchan, S. Fugkeaw

An auditor proves credentials, scope and policy freshness **without revealing them**; replay-safe, load-aware verifier shards and hierarchical Merkle log commitments (Rust, Amazon EC2). With batching: **1,415 proofs/s per core**, 2.31× zk-promises.

Submitted

FEDAC: A Federated Dynamic Access Control Framework for Cloud-Native Systems with Privacy-Preserving Authorization and Verifiable Governance Submitted · IEEE TDSC 2026

S. Junchan, K. Jairak, P. Pruksorranan, S. Fugkeaw

A control hub verifies each caller once and issues a short-lived, service-scoped capability; a sidecar beside every workload enforces it locally, so no network call sits on the request path. Four risk-adaptive authentication tiers up to a Groth16 zero-knowledge proof; blockchain-anchored audit. On AWS EKS against three re-implemented frameworks (Heimdall, ZKP-CapBAC, Atalay): **1.4 ms median latency**, about **7,100 requests/s**.

PQ-SCVA-MCK: Scope-Complete Verifiable Authorization and Impact-Bounded State Continuity for Post-Quantum Secure Multi-Cluster Kubernetes Submitted

S. Junchan, S. Fugkeaw

ML-DSA-signed proofs, anchored on Hyperledger Fabric, show that every deny rule was considered. About **28,000 authorizations/s** on Amazon EKS, 14.5–81.6× the throughput of three earlier schemes.

SWARM-Fog: Adaptive Swarm Intelligence for Real-Time Load Sharing and Fault Recovery in Large-Scale IIoT Fog-Cloud Systems Submitted

S. Junchan, S. Fugkeaw

Self-healing industrial IoT with no central controller; fog clusters schedule by trust. **100% recovery completeness** at 5–50% node failures, about **3–6× faster** recovery than completeness-matched baselines.

In preparation (first author, with Prof. S. Fugkeaw)

- **ZK-ALBA**: Authorization-Bound Zero-Knowledge and Load-Sharded Auditing for Cloud Logs (journal): integrity-check cost up to **12× lower** and preparation cost 13–112× lower than prior schemes.
- **FRESCO**: Freshness-Aware Elastic Orchestration of Stateful Services for the Multi-Cloud Continuum. Sparse-Merkle-tree freshness proofs steer scheduling and failover away from stale replicas; on AWS EKS, **0% freshness violations** where baselines reach 65–67%.
- **MoCA** (journal): AI agents that stay safe even when prompt injection succeeds. A hijacked agent cannot exceed its **signed human delegation**, checked at an MCP gateway and at the resource.
- **Threshold-gated encrypted SIEM**: a SIEM that works on **encrypted logs** shared by several organisations; tracing across them opens only when enough of them report the same signal.

Experience

Team Lead & Developer, TU Alert 2026 – Present
Thammasat University (client project) *Launching soon*

- **Lead a two-person student team** (we both code) hired by the university to build a LINE (LIFF) incident-reporting system that **routes each report to the office that owns it**. TypeScript (Hono, Bun, Prisma, Next.js); second-step staff sign-in, data retention and erasure tools, **Semgrep** and **gitleaks** in CI.

Information Security Intern (summer training, CSS300) Jun – Jul 2026
Allianz Ayudhya *Bangkok, Thailand*

- **Built two AI-agent prototypes for third-party risk management**: a questionnaire agent that answers due-diligence questionnaires with evidence, a reason and a confidence score, plus gap analysis, and a guidance agent that walks business owners through each step. Tested on real, anonymized cases.
- **Learned information security governance and risk management in practice** from my mentor, the CISO.

R&D Intern, Amity Solutions – *LLM chatbots end to end, knowledge base, chatbot pipeline* Jul – Aug 2023

- **Read recent LLM research papers, applied their ideas and tested them end to end**; built chatbots that answer from a knowledge base with retrieval-augmented generation (RAG): search the documents, give the best matches and the question to the LLM, return its answer.

Teaching and Mentoring

Founder, Curriculum Author & Lead Instructor 2022 – Present
TOI Tutor and ABIG Center (online) *toitutor.in.th*

- **Founded and lead two online olympiad-coaching platforms** (TOI, POSN) for hundreds of students.
- **Wrote the courses**: COM300 (C++ and algorithms: 9 chapters, about 270 pages, 128 recorded lessons) and COM100 (mathematics and Python: 10 chapters with answer keys).

Honours and Awards

- **Best Paper Award**, IEEE JCSSE 2026, for U-HAP · **National Champion, 1st Place & Gold Medal**, Thailand Olympiad in Informatics (TOI18), 2022.
- **1st Place Overall**, Image Recognition Programming Challenge (Osaka Institute of Technology and SIIT), 2025.
- **Full scholarship**, 4-Year Continuing Academic Outstanding Student Program, SIIT, 2023 – 2027.
- **Bronze Medal & Popular Vote**, SCIUS Forum 2022 (IoT cat feeder) · **Winner**, HotelDirect.in.th startup, depa Thailand Digital BIG BANG Regional, 2018 · **Bronze Medal**, National Robot Competition, 2017.

Certifications and Security Training

- **ISC2 Certified in Cybersecurity (CC)**, 2025 · **CompTIA Security+** (exam planned Dec 2026) · DataCamp: Azure and AWS, 2025.
- **SOSECURE Road to Cybersecurity 2025 (GEN6)**: eight modules, from networking to SOC.
- **NCSA CTF Thailand Tournament 2026** (Aug 2026), team member · TryHackMe and Hack The Box.

Skills

Cryptography:	zk-SNARKs (Groth16), Merkle commitments, post-quantum ML-DSA and ML-KEM
Cloud & Systems:	Kubernetes (RBAC, ABAC, webhooks), AWS (EKS, EC2), Docker, Hyperledger Fabric, OPA/Rego
Security Tools:	Nmap, Wireshark, Burp Suite, Metasploit, Splunk and Wazuh (SOC labs), OWASP Top 10
Programming:	**Python** and **C++** (main); also Go, Rust, TypeScript, C, Java, Bash, SQL
Languages:	Thai (native), English (fluent), Chinese (basic)

Leadership and Service

- **President, Sub Student Committee 32, SIIT** (2024 – present): lead a committee that represents the student body; organised 11 student events and led SIIT First Meet 2024 (about 500 participants, 100 staff).
- President, Psychology Club, Darunsikkhalai Science School, 2021.

Referees

Prof. Somchart Fugkeaw – research supervisor; co-author of all the papers above
School of ICT, SIIT, Thammasat University *Contact details on request*